

- 3 (a) From point of release to point of maximum compression,

Loss in gravitational potential energy = Gain in elastic potential energy

$$mg(0.10 + x_{\max}) = \frac{1}{2} kx_{\max}^2$$

$$0.30(9.81)(0.10 + x_{\max}) = \frac{1}{2}(80)x_{\max}^2$$

1

$$x_{\max} = 0.13 \text{ m}$$

1

- (b) Block continues increasing its kinetic energy until net force on it = 0.

When $mg = kx$

$$x = \frac{mg}{k} = \frac{0.30(9.81)}{80} = 0.0368 \text{ m}$$

1

Loss in GPE = Gain in EPE + Gain in KE

$$mg(0.10 + 0.0368) = \frac{1}{2} k(0.0368)^2 + \text{Gain in KE}$$

$$(0.30)(9.81)(0.10 + 0.0368) = \frac{1}{2}(80)(0.0368)^2 + \text{Gain in KE}$$

1

$$\text{Gain in KE} = 0.35 \text{ J}$$

1

- (c)

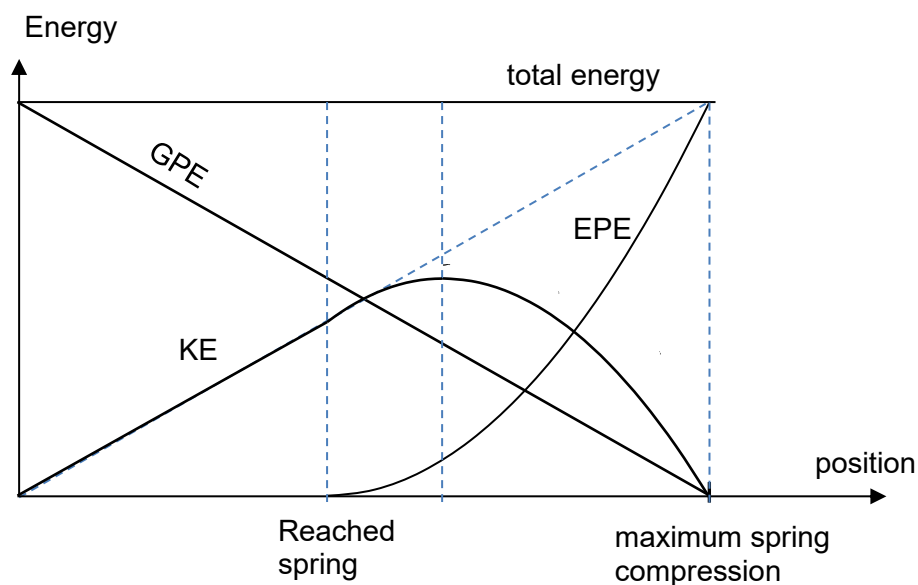


Fig. 3.2

1 mark for each graph.

[Total: 8]

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

4. (a) (i) The electric field strength E at a point is the **electric force per unit positive** 1